

1. (10分)

(a) 求解下面方程组。

(b) 求出方程组增广矩阵列向量组的一组极大无关组。

$$\begin{cases} x_1 + 3x_2 + x_3 = 2 \\ 3x_1 + 4x_2 + 2x_3 = 9 \\ -x_1 - 5x_2 + 4x_3 = 10 \\ 2x_1 + 7x_2 + x_3 = 1 \end{cases}$$

$$\begin{aligned} (a) \quad & \begin{bmatrix} 1 & 3 & 1 & 2 \\ 3 & 4 & 2 & 9 \\ -1 & -5 & 4 & 10 \\ 2 & 7 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & -5 & -1 & 3 \\ 0 & -2 & 5 & 12 \\ 0 & 1 & -1 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 3 & 6 \\ 0 & 0 & -6 & -12 \end{bmatrix} \\ & \rightarrow \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \therefore \begin{cases} x_1 = 3 \\ x_2 = 1 \\ x_3 = 2 \end{cases} \end{aligned}$$

(b) 易知 $|A|=0$, $\therefore \text{rank} < 4$

$$\text{而} \quad \begin{vmatrix} 3 & 4 & 2 \\ -1 & -5 & 4 \\ 2 & 7 & 1 \end{vmatrix} = \begin{vmatrix} 0 & -11 & 14 \\ -1 & -5 & 4 \\ 0 & -3 & 9 \end{vmatrix} = \begin{vmatrix} -11 & 14 \\ -3 & 9 \end{vmatrix} = -99 + 42 \neq 0$$

$\therefore$ 延伸组线性无关

$\therefore$ 前三列构成^{极大}线性无关组

2. (15分) 求下面行列式

a、

$$\begin{vmatrix} 1 & 2 & 3 & x \\ 2 & 3 & x & 1 \\ 3 & x & 1 & 2 \\ x & 1 & 2 & 3 \end{vmatrix}$$

b、

$$\begin{vmatrix} x_1 - a_1 & x_2 & x_3 & \cdots & x_n \\ x_1 & x_2 - a_2 & x_3 & \cdots & x_n \\ x_1 & x_2 & x_3 - a_3 & \cdots & x_n \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_1 & x_2 & x_3 & \cdots & x_n - a_n \end{vmatrix}$$

$$\begin{aligned} a \quad \text{原式} &= (6+x) \begin{vmatrix} 1 & 2 & 3 & x \\ 1 & 3 & x & 1 \\ 1 & x & 1 & 2 \\ 1 & 1 & 2 & 3 \end{vmatrix} = (6+x) \begin{vmatrix} 1 & 2 & 3 & x \\ 0 & 1 & x-3 & 1-x \\ 0 & x-3 & 1-x & 1 \\ 0 & 1-x & 1 & 1 \end{vmatrix} \\ &= (6+x) \cdot \begin{vmatrix} 1 & x-3 & 1-x \\ x-3 & 1-x & 1 \\ 1-x & 1 & 1 \end{vmatrix} \end{aligned}$$

$$\begin{vmatrix} 1 & x-3 & 1-x \\ x-3 & 1-x & 1 \\ 1-x & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1-x & 1 \\ 1 & 1 \end{vmatrix} - (x-3) \begin{vmatrix} x-3 & 1-x \\ 1 & 1 \end{vmatrix} + (1-x) \begin{vmatrix} x-3 & 1-x \\ 1-x & 1 \end{vmatrix}$$

$$= -x - (x-3) \cdot (2x-4) + (1-x) \cdot (-x^2+3x-4)$$

$$= -x - (2x^2-10x+12) + (x^3-3x^2+4x-x^2+3x-4)$$

$$(-x^3-4x^2+7x-4)$$

$$= x^3 - 6x^2 + 16x - 16$$

$$\therefore \text{原式} = (x+6) \cdot (x^3 - 6x^2 + 16x - 16)$$

$$b. \text{原式} = \begin{vmatrix} 1 & x_1 & x_2 & \dots & x_n \\ 0 & x_1-a & x_2 & \dots & x_n \\ 0 & x_1 & x_2-a & \dots & x_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & x_1 & x_2 & \dots & x_n-a \end{vmatrix}$$

$$= \begin{vmatrix} 1 & x_1 & x_2 & \dots & x_n \\ -1 & -a_1 & 0 & \dots & 0 \\ -1 & 0 & -a_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & 0 & 0 & \dots & -a_n \end{vmatrix}$$

$$= a_1 a_2 \dots a_n \begin{vmatrix} 1 & \frac{x_1}{a_1} & \frac{x_2}{a_2} & \dots & \frac{x_n}{a_n} \\ -1 & 1 & 0 & \dots & 0 \\ -1 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & 0 & 0 & \dots & 1 \end{vmatrix} = \begin{vmatrix} 1 - \sum_{i=1}^n \frac{x_i}{a_i} & \frac{x_1}{a_1} & \frac{x_2}{a_2} & \dots & \frac{x_n}{a_n} \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{vmatrix}$$

$$= (-1)^n \cdot \left(1 - \sum_{i=1}^n \frac{x_i}{a_i}\right) \cdot \prod_{i=1}^n a_i$$

3. (20分) 设行列式

$$\begin{vmatrix} \zeta_{11} & \zeta_{21} & \cdots & \zeta_{n1} \\ \zeta_{12} & \zeta_{22} & \cdots & \zeta_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ \zeta_{1n} & \zeta_{2n} & \cdots & \zeta_{nn} \end{vmatrix}$$

不为零。

(a)、证明满足下面方程组的未知量 x_{ijl} 唯一

$$\zeta_{jt}\zeta_{lt} = \sum_{i=1}^n x_{ijl}\zeta_{it}, 1 \leq j, l, t \leq n.$$

(b)、求出 x_{ijl} 的值, $1 \leq i, j, l \leq n$.

(a) 即: $\zeta_{jt}\zeta_{lt}$ 用 $\zeta_{1t}, \zeta_{2t}, \dots, \zeta_{nt}$ 的表示方式唯一

而后者是与 ζ_{lt}, ζ_{jt} 同一列的

$$\therefore \zeta_{jt}, \zeta_{lt} \in \alpha_t = (\zeta_{1t}, \zeta_{2t}, \dots, \zeta_{nt})$$

若表示方式不唯一, 即有 $x'_{ijl}, x'_{2jl}, \dots, x'_{njl}$ 与 $x_{ijl}, x_{2jl}, \dots, x_{njl}$ 不全相同

$$\text{作差得 } \sum_{i=1}^n (x_{ijl} - x'_{ijl}) \zeta_{it} = 0$$

而 $x_{ijl} - x'_{ijl}$ 不全为0

则 $\zeta_{1t}, \zeta_{2t}, \dots, \zeta_{nt}$ 线性相关

由于 j, l 可任取, 则推出行列式列向量组线性相关

则行列式为0, 矛盾

则未知量唯一

(b) $t=1, 2, \dots, n$ 时, 均有 $\zeta_{jt}\zeta_{lt} = \sum_{i=1}^n x_{ijl}\zeta_{it}$, 不影响 x 的取值

$$\therefore \text{对一组 } x, \text{ 有 } n \text{ 元线性方程组 } \begin{cases} \sum_{i=1}^n x_{ijl}\zeta_{i1} = \zeta_{j1}\zeta_{l1} \\ \vdots \\ \sum_{i=1}^n x_{ijl}\zeta_{in} = \zeta_{jn}\zeta_{ln} \end{cases}$$

$$\text{即: } \begin{bmatrix} \zeta_{11} & \zeta_{21} & \cdots & \zeta_{n1} \\ \zeta_{12} & \zeta_{22} & \cdots & \zeta_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ \zeta_{1n} & \zeta_{2n} & \cdots & \zeta_{nn} \end{bmatrix} \begin{bmatrix} x_{1jl} \\ x_{2jl} \\ \vdots \\ x_{njl} \end{bmatrix} = \begin{bmatrix} \zeta_{j1}\zeta_{l1} \\ \zeta_{j2}\zeta_{l2} \\ \vdots \\ \zeta_{jn}\zeta_{ln} \end{bmatrix}$$

由题设系数行列式 $|A| \neq 0$, 有唯一解

由克莱姆法则 记系数行列式为 $|A|$, B_i 表示系数行列式第 i 列换为

$$(\zeta_{j1}\zeta_{l1}, \zeta_{j2}\zeta_{l2}, \dots, \zeta_{jn}\zeta_{ln})^T$$

则有 $x_{ijl} = \frac{|B_l|}{|A|}$

4. (15分) 令矩阵

$$A = \begin{pmatrix} 1 & a_{12} & a_{13} \\ 0 & 1 & a_{23} \\ 0 & 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & b_{12} & b_{13} \\ 0 & 1 & b_{23} \\ 0 & 0 & 1 \end{pmatrix}.$$

1、求矩阵 A, B 的代数余子式 $A_{ij}, B_{ij}, 1 \leq i, j \leq 3$.

2、定义新矩阵

$$A^* = \begin{pmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{pmatrix}, B^* = \begin{pmatrix} B_{11} & B_{21} & B_{31} \\ B_{12} & B_{22} & B_{32} \\ B_{13} & B_{23} & B_{33} \end{pmatrix}.$$

求: A^*B^*AB 和 $B^*B^*A^*BABA^*B^*AB$.

4. $|A| = |B| = 1, \therefore \sum_{i=1}^3 a_{ij} \cdot A_{ij} = 1, \sum_{i=1}^3 a_{ij} \cdot A_{ik} = 0, j \neq k.$

$\therefore A_{33} = 1 = A_{11}, A_{23} = 0 = A_{13}.$

$$A^* = \begin{bmatrix} 1 & 0 & 0 \\ -a_{12} & 1 & 0 \\ a_{12}a_{23}-a_{13} & -a_{23} & 1 \end{bmatrix} \quad B^* = \begin{bmatrix} 1 & 0 & 0 \\ -b_{12} & 1 & 0 \\ b_{12}b_{23}-b_{13} & -b_{23} & 1 \end{bmatrix}$$

5. (15分) 设 n 为正整数, 数域 K 上 n 元非齐次线性方程组 $AX = \beta (\beta \neq 0)$ 有解, 且其系数矩阵 A 的秩满足 $1 \leq r(A) < n$, 证明: 必存在 $n - r(A) + 1$ 个线性无关的解向量 $\gamma_1, \gamma_2, \dots, \gamma_{n-r(A)+1}$, 使得方程组 $AX = \beta$ 的每个解都是这 $n - r(A) + 1$ 个解向量的线性组合.

对于齐次方程组 $AX = 0$, 其解空间维数为 $n - r(A)$, 记为 $\gamma_1, \gamma_2, \dots, \gamma_{n-r(A)}$

而 $AX = \beta$ 的解即 $AX = 0$ 的解加上一个特解

记有特解 $A\alpha = \beta$.

则 $AX = \beta$ 的解均可由 $\gamma_1, \gamma_2, \dots, \gamma_{n-r(A)}, \alpha$ 线性表出

(若有 $\alpha', A\alpha' = \beta$, 则 $A(\alpha - \alpha') = 0$, $\alpha - \alpha'$ 是 $AX = 0$ 的解, 可由 $\gamma_1, \gamma_2, \dots, \gamma_{n-r(A)}$ 线性表出)

只要证它们线性无关

若它们线性相关, 则有 α 可由 $\gamma_1, \gamma_2, \dots, \gamma_{n-r(A)}$ 线性表出

$\therefore$ 必有 $A\alpha = 0, \beta = 0$, 矛盾

$\alpha \in D$.

6. (15分) 设 A 为 3×2 矩阵, B 为 2×3 矩阵, 已知

$$AB = \begin{bmatrix} 8 & 2 & -2 \\ 2 & 5 & 4 \\ -2 & 4 & 5 \end{bmatrix}$$

a、求矩阵 A, B 的秩。

b、求 BA 。

$$6. (a.) \quad AB = \begin{bmatrix} 0 & 18 & 18 \\ 0 & 9 & 9 \\ -2 & 4 & 5 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \text{rank}(AB) = 2$$

$$\text{而 } \text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$$

$$\therefore \text{rank}(A) \geq 2, \text{rank}(B) \geq 2$$

$$\therefore \text{rank}(A) = \text{rank}(B) = 2.$$

(b)

7. (10分) 定义正整数集合 $\mathbb{Z}^+$ 上取值于数域 k 的函数 $f_1, \dots, f_r$. 证明: 不存在一组非零的数 $\ell_1, \dots, \ell_r$ 使得 $\ell_1 f_1 + \ell_2 f_2 + \dots + \ell_r f_r$ 为零函数的充要条件是存在 $i_1, i_2, \dots, i_r \in \mathbb{Z}^+$ 使得下面矩阵的秩为 r

$$\begin{pmatrix} f_1(i_1) & f_1(i_2) & \cdots & f_1(i_r) \\ f_2(i_1) & f_2(i_2) & \cdots & f_2(i_r) \\ \vdots & \vdots & \ddots & \vdots \\ f_r(i_1) & f_r(i_2) & \cdots & f_r(i_r) \end{pmatrix}.$$

① 充分性: 若存在一组非零数, 使 $\ell_1 f_1 + \ell_2 f_2 + \dots + \ell_r f_r = 0$

则对齐次方程 $A^T X = 0$, 存在一个非零解 $(\ell_1, \ell_2, \dots, \ell_r)$

与 $\text{rank}(A) = r$ 矛盾

② 必要性:

$$\text{设 } A_n = \begin{pmatrix} f_1(1) & f_1(2) & \cdots & f_1(n) \\ f_2(1) & f_2(2) & \cdots & f_2(n) \\ \vdots & \vdots & \ddots & \vdots \\ f_r(1) & f_r(2) & \cdots & f_r(n) \end{pmatrix}$$

则 $\text{rank}(A_n) = r_n \leq r$, 且 $r_n \leq r_{n+1}$, $\therefore r_n$ 存在最大值

即 $\exists n_0$, 当 $n \geq n_0$ 时, $r_n = M \leq r$.

若 $M < r$, 此时取 A_{n_0} 列向量组极大线性无关组 $\eta_1, \eta_2, \dots, \eta_M$

对方程 $\begin{bmatrix} \eta_1^T \\ \eta_2^T \\ \vdots \\ \eta_m^T \end{bmatrix} x = 0$, 由于 $M < r$, 则一定有非零解

取一个非零解 l .

则有 $\eta_1 \cdot l = 0, \eta_2 \cdot l = 0, \dots, \eta_m \cdot l = 0$

由于 $\eta_1, \eta_2, \dots, \eta_m$ 是极大线性无关组

$\therefore$ 任意一列均可由其线性表出

则有 $\beta_j = \sum_{k=1}^m c_k \eta_k, \forall j \in N$.

l 也是 β_j 的解

$\therefore l_1 f_1(j) + l_2 f_2(j) + \dots + l_r f_r(j) = 0, \forall j \in N$

与条件矛盾

$\therefore M = r$

$$\textcircled{1} \begin{vmatrix} 1+x_1^2 & x_1x_2 & \cdots & x_1x_n \\ x_2x_1 & 1+x_2^2 & \cdots & x_2x_n \\ \vdots & \vdots & \ddots & \vdots \\ x_nx_1 & x_nx_2 & \cdots & 1+x_n^2 \end{vmatrix} = \begin{vmatrix} 1 & \cdots & x_1 & x_2 & \cdots & x_n \\ 0 & 1+x_1^2 & x_1x_2 & \cdots & x_1x_n \\ 0 & x_2x_1 & 1+x_2^2 & \cdots & x_2x_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & x_nx_1 & x_nx_2 & \cdots & 1+x_n^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & x_1 & x_2 & \cdots & x_n \\ -x_1 & 1 & 0 & \cdots & 0 \\ -x_2 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -x_n & 0 & 0 & \cdots & 1 \end{vmatrix}, \text{ 记作 } D_{n+1}$$

$$\therefore D_{n+1} = D_n + (-1)^{n+2} \cdot x_n$$

$$\begin{vmatrix} -x_1 & 1 \\ -x_2 & 0 & 1 \\ \vdots & \vdots & \vdots \\ -x_n & 0 & 0 & \cdots & 0 \end{vmatrix}$$

$$= D_n + x_n^2$$

$$\therefore D_n = x_{n-1}^2 + D_{n-1}, \text{ 依此类推,}$$

$$D_n = \sum_{i=1}^n x_i^2 + D_1 = \sum_{i=1}^n x_i^2 + 1$$

$$\textcircled{2} \text{ 令 } e_1 = (0, 0, \cdots, 1, \cdots, 0)^T, K = (x_1, x_2, \cdots, x_n)^T$$

$$\text{则 } |A| = \det(e_1 + x_1K, e_2 + x_2K, \cdots, e_n + x_nK)$$

由于两列成比例, 行列式为0

$$\therefore |A| = \det(e_1, e_2, \cdots, e_n) + \sum_{i=1}^n (e_1, e_2, \cdots, x_iK, e_{i+1}, \cdots, e_n)$$

$$= \begin{vmatrix} 1 & & & & & \\ & 1 & & & & \\ & & \ddots & & & \\ & & & x_i^2 & & \\ & & & \vdots & & \\ & & & x_nx_i & \cdots & 1 \end{vmatrix} = x_i^2$$

